

# NOTICE

## Robotics & AI — Pre-Selection Test Syllabus | Class X

This is to inform all students that a Pre-Selection Test for Robotics & Artificial Intelligence will be conducted at the end of June. The syllabus covers all topics from April through June.

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Revision of Class IX &amp; New Age Robotic Systems</b> | <ul style="list-style-type: none"> <li>▶ Quick recap of Class IX Robotics &amp; AI (Python part)</li> <li>▶ Warehouse Robots, Assistant Robots, Smart Homes / Schools / Mobility</li> <li>▶ Autonomous Cars, Drones, Robots in Medicine</li> <li>▶ NARS relevance</li> </ul>                                                                                                                            |
| <b>From Robots to Cobots</b>                              | <ul style="list-style-type: none"> <li>▶ Difference between machine and robot (sensing, speed, task)</li> <li>▶ Cobots — definition, difference from robots, importance</li> <li>▶ Progress from Robots to Cobots</li> </ul>                                                                                                                                                                            |
| <b>Python — Strings</b>                                   | <ul style="list-style-type: none"> <li>▶ Introduction to strings; creating and traversing a string</li> <li>▶ Multiline strings, concatenation, escape sequences, string operators</li> <li>▶ In-built functions: len(), upper(), lower(), endswith(), startswith(), find(), islower(), replace(), count(), isalpha(), isdigit(), isalnum(), join()</li> <li>▶ Slicing: [start : end : step]</li> </ul> |
| <b>Actuators &amp; Controller for a Robotic System</b>    | <ul style="list-style-type: none"> <li>▶ Actuators: types (linear and rotary), joints and wheels</li> <li>▶ Controller: manual vs automatic control; control systems from daily life</li> <li>▶ Block diagram: Input → Controller → Robot → Feedback</li> <li>▶ Integrating sensors, actuators and controller</li> </ul>                                                                                |
| <b>Components of Robots — Gears &amp; Sensors</b>         | <ul style="list-style-type: none"> <li>▶ Gears: transmission, amplification of force, principles, types, gear ratio</li> <li>▶ Sensors: types by application (vision, tactile, temperature, range, proximity, motion, speech)</li> <li>▶ Internal vs external sensors; functions of sensors</li> </ul>                                                                                                  |
| <b>Python — Lists and Tuples</b>                          | <ul style="list-style-type: none"> <li>▶ Introduction to lists; creation; accessing elements</li> <li>▶ List operations: append, insert, extend, sort, search</li> <li>▶ Tuples: creation, accessing, deleting; converting list ⇌ tuple</li> </ul>                                                                                                                                                      |

**Best of luck, Class XI!**